



Alexandru Dabu

📍 Bucharest, Romania
☎ +40 756 517 830
✉ alexandru.dabu123@gmail.com
🌐 linkedin.com/in/alexandru-dabu
🐙 github.com/AnduDabu
🌐 Personal Website: www.alexandrudabu.com
🗣 Romanian — native ▪ English — fluent

📖 Summary

Graduate in **Automation and Computer Engineering**, currently working as a **Python developer within an actuarial team**, building the internal software behind **IFRS 17** and **Solvency II** reporting. I turn manual, repetitive closing processes into reliable automated workflows, from data consolidation and Excel reporting to cloud-orchestrated financial model runs. Alongside this, I freelance on full-stack web applications and automation for small businesses. Complementary hands-on experience ranging from hardware, low-level programming, and robotics to advanced control design, AI/ML-driven systems, and cross-platform mobile development. Passionate about exploring new technologies and applying technical skills to solve real-world challenges.

🎓 Education

2025–Present: 🎓 Master's Degree in Information Systems and Digital Transformation in Materials Processing, **University Politehnica of Bucharest**, Faculty of Materials Science and Engineering

2021–2025: 🎓 Bachelor's Degree in Automation and Computer Engineering, **University Politehnica of Bucharest**, Faculty of Automation and Computer Engineering

2017–2021: 🎓 High School Diploma, **Colegiul Național „Radu Greceanu”**, Slatina — Mathematics & Informatics

💼 Work Experience

🏢 **Allianz Services — Actuarial Python Developer** Dec 2025 – Present

Developer within the **Life & Health** actuarial team delivering **IFRS 17** and **Solvency II** reporting for Allianz entities in France, where I design, build and maintain the Python tooling, data pipelines and cloud workflows that replace manual work in the quarterly closing process.

- 🔹 **Automated the quarterly liability analysis** with a Python tool that ingests, cleans and reconciles data from multiple source files and produces a detailed multi-sheet Excel report on liabilities and key indicators (BEL, PVFCF), removing a recurring manual effort from every closing.
- 🔹 **Cut a three-day manual setup to about 10 minutes** with a **Workspace Creation Tool** that provisions the full quarterly working environment for the team — a **400+ folder** structure carrying over the required files from the previous quarter — then ported it from a local script to **Unify** (Willis Towers Watson's cloud orchestration platform) so any colleague can trigger it on demand.
- 🔹 **Saved roughly 43 hours of setup and compute time per closing** by moving financial model execution to the cloud with a second Unify workflow that runs **deterministic and stochastic projections** over our portfolio and economic scenarios, replacing the manual cycle of opening, compiling and parameterizing it in **Risk Agility FM**.
- 🔹 **Automated post-run log analysis** in the same workflow, parsing raw model output logs to extract, clean and consolidate the errors and skipped points of every scenario into a single readable summary, previously gathered by hand.
- 🔹 **Owned the tools end to end**, translating requirements from actuaries into production software, writing Excel automations in **Python** while migrating legacy **VBA** macros to it, documenting the tools and processes so the team can maintain them, and running delivery in **Agile** sprints with Unify migrations and tickets tracked in **Jira/Confluence**.
- 🔹 **Involved in the quarterly closing process on IFRS 17 and Solvency II standards**, preparing inputs and performing deterministic and stochastic forecasts, executed end-to-end through the Unify workflow I built.
- 🔹 **Analyzed the evolution of key indicators** — BEL (SII), PVFCF (IFRS 17), reserves and market value — under different economic scenarios, shocks and asset assumptions.

🔧 Tech Stack: Python (Pandas, NumPy, openpyxl), VBA, PowerShell, Excel Automation, WTW Unify, Risk Agility FM (RAFM), Data Cleaning & Analysis, UML, Agile, Jira/Confluence, Git, IFRS 17, Solvency II

🏠 **Carpasoft — Mobile Developer Intern**

Jul 2024 – Oct 2024

The internship focused on building a cross-platform mobile application for pet-related management services.

- 🔹 **Extended user and pet profiles** with editable attributes and health records.
- 🔹 **Developed booking and scheduling features** for veterinary visits, grooming, and pet walking services.
- 🔹 **Improved community features** such as posts, stories, and real-time chat to connect users and providers.
- 🔹 **Integrated maps and geolocation** for discovering pet-related facilities and tracking lost pets.
- 🔹 **Improved software quality** by implementing automated test suites within CI/CD pipelines.

🔧 Tech Stack: Flutter, Google Cloud Services, Maps/Geolocation, GitLab, Linux, Android, iOS

Freelance

Developer & Technical Consultant

Independent work for local businesses, spanning web development, design for CNC metal fabrication, and the documentation behind residential solar grid connections.

- **Design and build websites for local businesses** — recent work at **alma-wood.ro** and **duiras.ro**. Written in **TypeScript** on **Next.js**, server-rendered and deployed to **Cloudflare's edge network**, with a motion layer built on **GSAP** and **Framer Motion**, and **Playwright** end-to-end tests before release. Optional **headless-CMS** integration lets owners edit their own content, and **SEO** plus **Google Business Profile** setup makes sure the client is actually found.
✂ Next.js, React, TypeScript, Tailwind CSS, shadcn/ui, GSAP, Framer Motion, Sharp, Playwright, Sanity CMS, Cloudflare Pages, SEO
- **Built and maintain an internal agency panel** that runs the business side of this work: **Google OAuth** sign-in, a serverless **Cloudflare D1** database behind a type-safe ORM, schema validation on every input, and scheduled background workers that issue recurring invoices, keep performance caches warm and watch third-party API quotas.
✂ Next.js, TypeScript, Drizzle ORM, Cloudflare D1, Cloudflare Workers, Google OAuth, Resend, Zod, Playwright
- **Design decorative metalwork for CNC laser cutting** — panels, gates, fences and signage — taking each piece from concept to cut-ready **DXF/SVG**. A geometry generator I built in Python handles the repetitive work: fitting patterns to the sheet, placing mounting tabs, and adding the small bridges that stop cut pieces falling through the laser bed.
- **Prepare and submit grid-connection documentation for solar prosumers**, including the **single-line electrical diagrams**, filed through the **DEO Connect** platform. I automated the repetitive work in Python — assembling each client's document set, placing signatures, and compressing bundles to fit the platform's upload limits.

Projects

Basketball Social Media & AI Shot Analysis App (B.Sc. Thesis)

2025

The project provided real-time computer vision for basketball shot evaluation, complemented by features that foster player interaction and feedback sharing.

- **Designed and implemented a cross-platform application** combining social networking with AI-powered basketball shot analysis.
 - **Enabled engagement features** including accounts, personal profiles, posts, comments, private messaging, and notifications.
 - **Delivered automated performance feedback** by building a YOLOv8-based shot detection pipeline with pose estimation and trajectory heuristics.
 - **Optimized inference pipeline** to run **real-time video analysis at 30 FPS on GPU**, with fallback to CPU for broader compatibility.
 - **Improved training usability** through interactive maps of basketball courts and an integrated chatbot for tactical advice.
- ✂ Tech Stack: Python(YOLOv8, OpenCV, Flask), Flutter, Firebase, REST, Google Maps API, OpenAI API

Path Planning in Radioactive Environment using Probabilistic Roadmaps

2025

The project focused on building a customizable, safe and efficient robot navigation in dangerous environments.

- **Built a path planning framework** using Probabilistic Roadmaps (PRM) for efficient navigation in complex, static obstacle environments.
 - **Designed interactive tools**, including a simulator and visualization interfaces, enabling experimentation and analysis of pathfinding behavior.
 - **Improved safety and robustness** by implementing obstacle inflation, safety margins, and adjustable connection thresholds, validated across diverse maps.
- ✂ Tech Stack: Python, NumPy, Tkinter, Matplotlib, PIL

Multi-Agent Formation Control

2025

The project aimed to create a system aimed at enabling coordinated movement of robot teams while preserving a stable formations.

- **Applied advanced control methods** by combining graph Laplacians, control barrier functions, and spiking neural networks.
 - **Increased robustness** by avoiding collisions through distributed control strategies and dynamically changing formations (line, square, diamond, trapezoid).
 - **Explored scalability and performance trade-offs** with trajectory visualizations, scalability and stability plots for effective evaluation.
- ✂ Tech Stack: Python, NumPy, SciPy, Matplotlib, PyTorch

Rain Prediction in Australia

2023

The project investigated how machine learning can leverage historical weather data to improve short-term rainfall predictions and model interpretability.

- **Achieved 80% prediction accuracy** by developing an end-to-end ML pipeline to forecast next-day rainfall using a dataset of 140k+ records.
 - **Improved model accuracy by 12%** compared to logistic regression baseline using Random Forests.
 - **Reduced false negatives by 18%** through hyperparameter tuning, improving recall for rainy days.
 - **Conducted exploratory data analysis** to identify high-impact meteorological features correlated with rainfall, strengthening model transparency and explainability.
- ✂ Tech Stack: Python, Pandas, scikit-learn, Matplotlib, Jupyter

ARP Spoofing & Man-in-the-Middle

2024

The project demonstrated network vulnerabilities by implementing ARP spoofing in a controlled lab environment and exploring defenses.

- **Developed packet manipulation routines** for interception, modification, and forwarding of packets.
- **Simulated ARP spoofing attacks** to show how attackers can intercept and manipulate network traffic.
- **Exposed security risks** such as credential theft, silent monitoring, and session hijacking.
- **Documented countermeasures** including encrypted protocols, static ARP entries, and inspection tools.
- **Strengthened networking expertise** through analysis of ARP/TCP traffic with Wireshark.

✂ Tech Stack: Python, Scapy, Wireshark, Kali Linux

Image Processing Pipeline in Verilog

2023

The project implemented a hardware image processing pipeline on FPGA using finite state machines and memory management techniques.

- **Implemented a hardware pipeline** performing flip, grayscale, and sharpen operations on FPGA.
- **Ensured synchronization** with FSMs, registers, and counters coordinating pixel processing cycles.
- **Balanced performance and resource use** with sequential and parallel logic optimized for throughput.

✂ Tech Stack: Verilog, FSM, FPGA tools, Xilinx

E-commerce Management System

2023

The project provided a prototype of an online store, from catalog management to checkout and returns, built with structured software engineering methods.

- **Delivered core features of an e-commerce platform**, including product catalog, cart management, and order lifecycle.
- **Driven team collaboration** with UML diagrams for architecture, user flows, and business processes.
- **Strengthened testing coverage** with unit tests for order, cart, and return functionality.

✂ Tech Stack: Java, Spring Boot (intro), JUnit, UML